



Modelling Financial Markets

Seminar 5: Conditional Volatility — GARCH Models



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ US: United States

Outline

Introduction

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Run the GARCH variance recursion and its forecasts step by step
- ▣ Estimate GARCH, GJR-GARCH and heavy-tailed variants and diagnose the residuals
- ▣ Compare models with likelihood ratios, information criteria and out-of-sample losses
- ▣ Measure estimation uncertainty with a parametric bootstrap

- ▣ Background: Lecture 5; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** five pen-and-paper exercises (A1–A5)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** one open question, a seed for the team project (C1), and a critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every estimate with its uncertainty: robust standard error, bootstrap interval or p -value
- Check every fitted model: convergence, parameters at a bound, standardised residuals

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (GARCH recursion and forecast, step by step), A3 (long-run variance and half-life)
- Part B: B1 (GARCH-t on the S&P 500 with diagnostics), B2 (GJR vs GARCH, LR test), B5 (QLIKE, Diebold–Mariano, Mincer–Zarnowitz), B7 (parametric bootstrap)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A5 (model: A1, A3)
- Part B: B3 (model: B2), B4 (model: B1, B2), B6 (model: B5), B8 (model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A3)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter5_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib` and `arch`
- Series: S&P 500, BET, Bitcoin; daily closes up to 18 September 2026
- Returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; equity indices on weekdays, Bitcoin 7 days a week
- Part C joins the BET and S&P 500 prices on common days first, then computes returns

Formulas Used Today

- GARCH(1,1) [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; GJR [Glosten, Jagannathan & Runkle \(1993\)](#): add $\gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2$
- Long-run variance $\sigma^2 = \omega / (1 - \alpha - \beta)$; forecast $\mathbb{E}_t \sigma_{t+h}^2 = \sigma^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \sigma^2)$
- Half-life $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$; likelihood ratio $LR = 2(\ell_1 - \ell_0) \sim \chi_m^2$
- QLIKE loss $r_t^2 / h_t + \ln h_t$ [Patton \(2011\)](#); Diebold–Mariano t on the loss differential [Diebold & Mariano \(1995\)](#)
- Sign-bias regression $\hat{z}_t^2$ on S_{t-1}^- , $S_{t-1}^- \hat{z}_{t-1}$, $S_{t-1}^+ \hat{z}_{t-1}$ [Engle & Ng \(1993\)](#)

A1: GARCH Recursion and Forecast, Step by Step [Solved]

Task

- ▣ GARCH(1,1) with $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$; start at $\sigma_1^2 = \omega / (1 - \alpha - \beta)$
- ▣ Shocks (%): $\varepsilon_1 = -2.0$, $\varepsilon_2 = 0.5$, $\varepsilon_3 = 3.0$, $\varepsilon_4 = -1.0$
- ▣ (a) $\sigma_2^2, \dots, \sigma_5^2$
- ▣ (b) Forecasts $\mathbb{E}_4 \sigma_{5+h}^2$ for $h = 0, \dots, 4$ and the 5-day variance
- ▣ (c) Half-life of a variance shock

Solution

- ▣ $\sigma_1^2 = 0.05 / 0.05 = 1.0000$
- ▣ $\sigma_2^2 = 0.05 + 0.4000 + 0.8500 = 1.3000$;
 $\sigma_3^2 = 0.05 + 0.0250 + 1.1050 = 1.1800$
- ▣ $\sigma_4^2 = 0.05 + 0.9000 + 1.0030 = 1.9530$;
 $\sigma_5^2 = 0.05 + 0.1000 + 1.6600 = 1.8100$
- ▣ (b) Gap to the long run 0.8101 shrinks by 0.95 per day:
 1.8100, 1.7695, 1.7311, 1.6945, 1.6598
 ► 5-day variance 8.665, volatility 2.944%
- ▣ (c) $\ln 0.5 / \ln 0.95 = 13.5$ days

A2: The Same Shocks in a GJR-GARCH [Proposed]

Solution discussed in class

Task

- GJR-GARCH(1,1): $\omega = 0.05$, $\alpha = 0.05$, $\gamma = 0.10$, $\beta = 0.85$; same shocks as A1
- (a) Persistence $\alpha + \gamma/2 + \beta$ and long-run variance, for symmetric z_t
- (b) $\sigma_2^2, \dots, \sigma_5^2$ starting from the long-run variance
- (c) Compare with A1 after the shocks -2 and $+3$: which model reacts more, and why?
- Model: A1

A3: Long-Run Volatility and Half-Life from Estimates [Solved]

Task

- GARCH(1,1)-t, daily returns in %, 2000–2026 (Lecture 5)
- S&P 500: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.1226$, $\hat{\beta} = 0.8721$
- (a) Derive $\sigma^2 = \omega / (1 - \alpha - \beta)$ from the recursion
- (b) Long-run daily and annual volatility (251.4 days per year)
- (c) Half-life; compare with the sample volatility 19.2%

Solution

- (a) Stationarity: $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2 = \sigma^2$ and $\mathbb{E}\varepsilon_{t-1}^2 = \sigma^2$, so $\sigma^2 = \omega + (\alpha + \beta)\sigma^2$
- (b) $\hat{\alpha} + \hat{\beta} = 0.9947$; $\hat{\sigma}^2 = 0.0157 / (1 - 0.9947) = 2.972$, i.e. 1.724% per day and 27.3% per year
- (c) $\ln 0.5 / (-0.00530) = 130.7$ days
- The long-run volatility exceeds the sample one: dividing by $1 - 0.9947$ amplifies small errors in $\hat{\alpha} + \hat{\beta}$

A4: Is the Fourth Moment Finite? [Proposed]

Solution discussed in class

Task

- With Normal z_t , $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$ (Seminar 1, A9)
- (a) S&P 500, GARCH-N: $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$; check the condition and compute K
- (b) Compare with the sample kurtosis 13.7
- (c) Student-t innovations with $\nu = 6.26$ have kurtosis $\kappa_z = 3(\nu - 2)/(\nu - 4)$; the condition becomes $(\alpha + \beta)^2 + (\kappa_z - 1)\alpha^2 < 1$. Check it for $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$
- Model: A3

A5: EWMA as a GARCH Special Case [Proposed]

Solution discussed in class

Task

- EWMA J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, $\lambda = 0.94$
- (a) Write σ_t^2 as a weighted sum of past r_{t-j}^2 ; give the weights of $j = 1, 10, 50$
- (b) Which share of the total weight falls on the last 20 days?
- (c) Which GARCH parameters give EWMA? Half-life of the weights; forecast for h days ahead
- Model: A1, A3

B1: GARCH-t on the S&P 500 with Diagnostics [Solved]

Task

- ▣ S&P 500, daily returns in %, 2000–2026
- ▣ (a) Test for ARCH effects: $Q(10)$ on r_t^2 and $LM(5)$
- ▣ (b) Fit GARCH(1,1) with Student-t innovations; report robust [Bollerslev & Wooldridge \(1992\)](#) and classical standard errors
- ▣ (c) Diagnose $\hat{z}_t$: $Q(10)$ on $\hat{z}_t$ and $\hat{z}_t^2$, $LM(5)$, sign-bias tests [Engle & Ng \(1993\)](#)
- ▣ Code: `arch_model(r, p=1, q=1, dist='t')`

Interpretation

- ▣ Which diagnostics pass and which fail?
- ▣ What does the failure suggest about the model?
- ▣ Why are the robust standard errors larger?

B1: Results [Solved]

	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\nu}$	$\hat{\alpha} + \hat{\beta}$
Estimate	0.0157	0.1226	0.8721	6.26	0.9947
Robust s.e.		0.0107	0.0103	0.48	
Classical s.e.		0.0101	0.0096		

- (a) Before the model: $Q(10)$ on $r_t^2 = 6133$, $LM(5) = 1679$: strong ARCH effects
- (c) After: $Q(10)$ on $\hat{z}_t^2 = 13.6$ ($p = 0.192$), $LM(5) = 5.8$ ($p = 0.326$): clustering removed
 - ▶ Sign bias $t = 2.47$, positive size bias $t = -2.56$, joint $\chi^2_3 = 39.3$: asymmetry left; try GJR (B2)
 - ▶ $Q(10)$ on $\hat{z}_t = 21.9$ ($p = 0.016$): small autocorrelation in the mean
- Robust errors are larger because z_t is not exactly Student-t (skewness -0.58)

B2: GJR vs GARCH on the S&P 500 [Solved]

Task

- (a) Fit GJR-GARCH(1,1)-t [Glosten, Jagannathan & Runkle \(1993\)](#); report $\hat{\gamma}$ with its robust t -statistic
- (b) LR test of $\gamma = 0$ against GARCH(1,1)-t (B1)
- (c) Compare AIC and BIC; repeat the sign-bias test on the GJR residuals

Interpretation

- Is the asymmetry economically large? Use the news impact curve
- Why is $\hat{\alpha}$ exactly 0? Is the LR test still valid?

B2: Results [Solved]

- (a) $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.03$), $\hat{\beta} = 0.881$
- (b) $\ell_{GARCH} = -9062.5$, $\ell_{GJR} = -8944.6$: $LR = 235.65$, $p < 0.001$
 - ▶ 5% critical value of χ^2_1 : 3.84; rejected by a wide margin
- (c) $\Delta AIC = -233.7$, $\Delta BIC = -226.8$ (GJR minus GARCH): all criteria prefer GJR; joint sign-bias χ^2_3 falls from 39.3 to 23.5
- $\hat{\alpha} = 0$: positive shocks do not raise volatility; the estimate is on the boundary, but $\gamma = 0$ is an interior restriction, so the LR test remains valid

B3: GJR vs GARCH on the BET [Proposed]

Solution discussed in class

Task

- BET, daily returns in %, 2000–2026
- (a) Fit GARCH(1,1)-t and GJR-GARCH(1,1)-t; report $\hat{\gamma}$ and its robust t
- (b) LR test of $\gamma = 0$; (c) AIC and BIC
- Model: B2

Interpretation

- Do the LR test and BIC agree? Why can they disagree?
- Is the leverage effect weaker on the BET than on the S&P 500? Give one economic reason

B4: Which Innovations for Bitcoin? [Proposed]

Solution discussed in class

Task

- Bitcoin, daily returns in %, 2014–2026 (365 days a year)
- (a) Fit GARCH(1,1) with Normal, Student-t, skewed-t Hansen (1994) and GED innovations; compare BIC
- (b) LR test of $\lambda = 0$ (skewed-t vs Student-t)
- (c) Report $\hat{\nu}$ and $\hat{\alpha} + \hat{\beta}$; any parameter at a bound?
- Model: B1, B2

Interpretation

- The sample skewness of $\hat{z}_t$ is negative; does the likelihood agree? Why not?
- What does $\hat{\nu} < 4$ imply for the kurtosis and for risk measures?

B5: Forecast Comparison on the S&P 500 [Solved]

Task

- Out of sample 2016–2026: re-estimate each January on all data since 2000; one-day-ahead forecasts h_t
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t, EWMA ($\lambda = 0.94$), Hist-20 (mean of the last 20 r_t^2)
- (a) Mean QLIKE [Patton \(2011\)](#); (b) Diebold–Mariano t against GARCH-t, HAC with 5 lags [Diebold & Mariano \(1995\)](#)
- (c) Mincer–Zarnowitz regression $r_t^2 = a + bh_t$ [Mincer & Zarnowitz \(1969\)](#), Wald test of $a = 0$, $b = 1$

Interpretation

- Which models are significantly worse than GARCH-t?
- Why is R^2 low even for good forecasts?

B5: Results [Solved]

Model	QLIKE	DM t	MZ b	MZ R^2	Wald p
GARCH-N	0.7510	-1.47	0.97	0.250	0.995
GARCH-t	0.7570	—	0.88	0.238	0.804
GJR-t	0.7319	-1.22	0.84	0.276	0.680
EGARCH-t	0.7454	-0.36	1.24	0.293	0.713
EWMA	0.8149	3.48	0.89	0.172	0.685
Hist-20	0.8888	4.10	0.68	0.152	0.028

- (a–b) 2692 days; lowest QLIKE: GJR-t; only EWMA and Hist-20 are significantly worse ($t = 3.48, 4.10$)
- (c) Unbiasedness not rejected for the GARCH models; rejected for Hist-20
- R^2 is low because $r_t^2 = \sigma_t^2 z_t^2$: the proxy is noisy even when $h_t = \sigma_t^2$

B6: Forecast Comparison on the BET [Proposed]

Solution discussed in class

Task

- Repeat B5 for the BET, same design and period (2016–2026)
- Report QLIKE, the Diebold–Mariano t against GARCH- t and the Mincer–Zarnowitz slope
- Model: B5

Interpretation

- Does the ranking of models change between the S&P 500 and the BET?
- Which part of the BET model would you improve first (recall the diagnostics of Lecture 5)?

B7: How Precise Is the Half-Life? A Parametric Bootstrap [Solved]

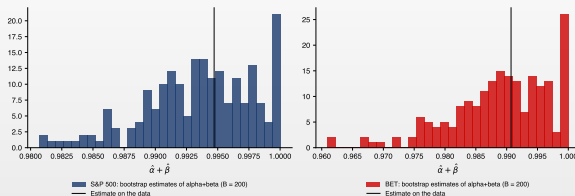
Task

- ▣ S&P 500, GARCH(1,1)-t fitted in B1
- ▣ (a) Simulate 200 paths of the same length from the fitted model
- ▣ (b) Re-estimate GARCH-t on each path; store $\hat{\alpha} + \hat{\beta}$ and the half-life
- ▣ (c) 95% percentile intervals; compare the bootstrap standard deviation with the robust (delta-method) standard error

Interpretation

- ▣ Why is the interval for the half-life so asymmetric?
- ▣ Can we reject IGARCH ($\alpha + \beta = 1$)?

B7: Results [Solved]



- Estimate 0.9947 (half-life 131 days); bootstrap 95% interval [0.9839; 1.0000]
- Bootstrap s.d. 0.0044 vs robust s.e. of $\hat{\alpha} + \hat{\beta}$ 0.0047: close
- Half-life: from 43 days to infinity; 5.5% of the replications hit $\alpha + \beta = 1$
 - $\ln 0.5 / \ln x$ explodes as $x \rightarrow 1$: a symmetric interval for x becomes a very skewed one for the half-life
- IGARCH cannot be rejected at 5%: the upper bound of the interval reaches 1

B8: Bootstrap for the BET [Proposed]

Solution discussed in class

Task

- Repeat B7 for the BET (GARCH(1,1)-t, 2000–2026), with the same number of replications
- Report the interval for $\alpha + \beta$ and for the half-life, and the share of replications with $\alpha + \beta = 1$
- Model: B7

Interpretation

- Is the BET half-life shorter than the S&P 500 one with confidence? What would a formal test need?

C1: Is the Romanian Market's Volatility Driven by the US? [Proposed]

Solution discussed in class

The question

- Wall Street closes after the Bucharest Stock Exchange: yesterday's S&P 500 shock is news for today's BET
- Does u_{t-1}^2 , the squared lagged S&P 500 return, raise the conditional variance of the BET?
- Model: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta u_{t-1}^2$, with $r_t = \mu + \phi u_{t-1} + \varepsilon_t$

Deliverable

- A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- Prices joined on common trading days first, then returns
- The likelihood written step by step (Student-t), estimated by numerical optimisation
- An LR test of $\delta = 0$; note that $\delta \geq 0$ puts the null on the boundary
- Stability: two subperiods (e.g. before and after 2013)
- The share of the BET variance attributable to US shocks, and how it changes over time

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily S&P 500 closes. Fit a GARCH(1,1) with Student- t innovations using arch and give me the annualised long-run volatility and the half-life of a shock.”

The assistant's answer

```
import numpy as np
from arch import arch_model
r = np.log(px).diff().dropna()          # daily log returns
am = arch_model(r, vol='GARCH', p=1, q=1, dist='t')
res = am.fit(displ='off')
om = res.params['omega']
a, b = res.params['alpha[1]'], res.params['beta[1]']
lr_var = om / (1 - a - b)               # long-run daily variance
ann_vol = lr_var * 252                  # annualised volatility
half_life = np.log(0.5) / np.log(b)    # half-life in days
```

- “For the S&P 500 this gives $\text{ann_vol} = 0.075$ (7.5% a year) and a half-life of 5.1 days.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs, but the answer contains **three** mistakes typical for GARCH work
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a warning, a number from A3, a slide of the lecture
 - ▶ Write the corrected line(s) and the corrected numbers
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: compare with A3 and with the slide “Common Pitfalls”

Common Pitfalls

- ▣ Fitting GARCH to returns in decimals: the optimiser may stop far from the maximum; use % or rescale
- ▣ Reading classical standard errors when the innovations are not Normal; use robust ones
- ▣ Trusting long-run volatility and half-life when $\alpha + \beta \approx 1$ without an interval
- ▣ Evaluating forecasts in sample, or with losses that are not robust to a noisy proxy
- ▣ Using an LR test for a parameter on the boundary (e.g. $\delta = 0$ with $\delta \geq 0$) with the usual χ^2 critical value
- ▣ Joining two markets on returns instead of prices: creates spurious multi-day returns

Take-Home Assignment (Team Project)

- ▣ Choose one series not covered in class (another index, an exchange rate, a commodity or a crypto-asset)
- ▣ Test for ARCH effects; fit GARCH, GJR and EGARCH with three innovation distributions; select by BIC and diagnostics
- ▣ Evaluate one-day and one-month forecasts out of sample against EWMA with QLIKE and Diebold–Mariano
- ▣ Report the half-life with a bootstrap interval; for an entropy-based alternative, see [Pele, Lazar & Dufour \(2017\)](#)
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Applied Time Series Analysis with Python

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_05

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